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|--------------|--------------------------------------|
| Started on   | Monday, 24 August 2026, 5:49 PM      |
| State        | Finished                             |
| Completed on | Thursday, 17 September 2026, 9:00 AM |
| Time taken   | 23 days 15 hours                     |
| Marks        | 1.00/1.00                            |
| Grade        | 10.00 out of 10.00 (100%)            |

**Question 1** | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int findFirstZero(int arr[], int low, int high) {
5      if (high >= low) {
6          int mid = low + (high - low) / 2;
7
8          if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
9              return mid;
10         }
11
12         if (arr[mid] == 1) {
13             return findFirstZero(arr, mid + 1, high);
14         }
15
16         return findFirstZero(arr, low, mid - 1);
17     }
18
19     return -1;
20 }
21
22 int main() {
23     int m;
24
25     if (scanf("%d", &m) != 1) {
26         return 0;
27     }
28
29     int *arr = (int*)malloc(m * sizeof(int));
30     if (arr == NULL) {
31         return 1;
32     }
33
34     for(int i = 0; i < m; i++) {
35         scanf("%d", &arr[i]);
36     }
37
38     int firstZeroIndex = findFirstZero(arr, 0, m - 1);
39
40     int zeroCount = (firstZeroIndex == -1) ? 0 : (m - firstZeroIndex);
41
42     printf("%d\n", zeroCount);
43
44     free(arr);
45
46     return 0;
47 }
```

|   | Input                                                                                             | Expected | Got |   |
|---|---------------------------------------------------------------------------------------------------|----------|-----|---|
| ✓ | 5<br>1<br>1<br>1<br>0<br>0                                                                        | 2        | 2   | ✓ |
| ✓ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1                                         | 0        | 0   | ✓ |
| ✓ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0                                                    | 8        | 8   | ✓ |
| ✓ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2        | 2   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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